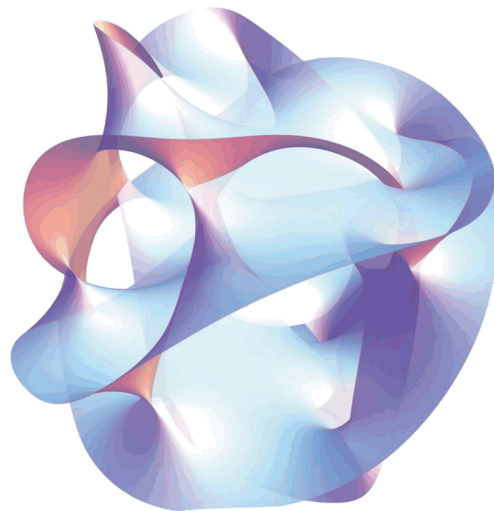


Dimensional reduction in supergravity and duality symmetries

Adam Franchot-Fowler

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Internship tutors : M. Camille Eloy

Referent teacher : M. Stéphane Douin

Éducational institution : Université Paris-Saclay / Magistère de physique fondamentale d'Orsay

Internship host company : Laboratoire de Physique (UMR CNRS 5672) ENS de Lyon

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Notations and abbreviations

We will during this internship use the vielbein formalism. We will use the same notation as in [1] which is

- Greek hatted letter for curved indices ($\hat{\mu}, \hat{\nu}, \hat{\rho}, \dots$)
- Latin hatted letter for flat indices ($\hat{p}, \hat{m}, \hat{n}, \dots$)
- For space-time indices taking D values, late greek letters are used in the curved case ($\mu, \nu, \rho, \dots$) and Latin letters in the flat case (p, m, n)
- For internal indices taking E values, early Greek letters in the curved case ($\alpha, \beta, \gamma, \dots$) and Latin letters in the flat case (a,b,c)

The general coordinate $x^{\hat{\mu}}$ will thereofre be made of space time coordinate x^{μ} and internal coordinate y^{α} . The flat lorenz

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1 Introduction

Since the 19th-century unification of electricity and magnetism, and the later development of classical and quantum field theory, physicists have been looking for a single mathematical framework that could describe all fundamental interactions at once. A lot of progress has been made for the electromagnetic, weak and strong forces, which are now successfully unified in the Standard Model. Gravity, however, remains the main stumbling block. Unlike the other forces, it resists quantization in a straightforward way, and its description by General Relativity [2] sits completely apart from the quantum field-theoretic language we use for everything else.

One of the earliest attempts to change this was proposed by Kaluza in 1921 [3] and later refined by Klein: by adding one extra spatial dimension compactified on a small circle, five-dimensional General Relativity naturally reduces to four-dimensional Einstein gravity coupled to electromagnetism. Even if the original theory had problems, this Kaluza-Klein mechanism opened a whole new way of thinking about unification, and it is now at the heart of modern approaches like supergravity [4] and string theory.

2 Theoretical tool

Here you will find some GR reminder, which hopefully will make it easier to understand the mathematical foundation on which this internship was build

2.1 spin connexion, vielbein and flatspace

In general relativity we often encounter difficult to handle metrics such as the Kerr-Newman metric. A good idea we can have is to then work in an orthonormal basis $\{e_\mu^a\}$, which follow:

$$g_{\mu\nu} V_a^\mu V_b^\nu = \eta_{ab} \quad (1)$$

where g is the original metric and η is the Minkowski metric.

We call this basis the vielbein. In this basis we will use the Minkowski metric to raise and lower indices. Of course we can define the dual of the vielbein otherwise called the inverse vielbein with

$$V_\nu^a V_b^\nu = \delta_b^a \quad (2)$$

and of course that mean that we can go back with to the original metric with

$$g_{\mu\nu} = \eta_{ab} V_\mu^a V_\nu^b \quad (3)$$

Of course we can also change the basis any random vector, form or tensor with this, the method is to simply do

$$T^a = V_\mu^a T^\mu \quad (4)$$

$$T_a = V_a^\mu T_\mu \quad (5)$$

$$T_b^a = V_\mu^a V_b^\nu T_\nu^\mu \quad (6)$$

Now in this internship we will not be in four dimensions, but rather in 4 time-space dimension, with E supplementarie space dimensions. And to pass from 4 + E total dimension to the other thing we use the vielbein in 4+E dimension the the 4 dimensions vielbein like this:

$$T^\mu = V_r^\mu \hat{V}_{\hat{\mu}}^r \hat{T}^{\hat{\mu}} \quad (7a)$$

$$T^\alpha = V_a^\alpha \hat{V}_{\hat{\mu}}^a \hat{T}^{\hat{\mu}} \quad (7b)$$

$$T_\mu = V_\mu^r \hat{V}_r^{\hat{\mu}} \hat{T}_{\hat{\mu}} \quad (7c)$$

$$T_\alpha = V_\alpha^a \hat{V}_a^{\hat{\mu}} \hat{T}_{\hat{\mu}} \quad (7d)$$

$$(7e)$$

Now of course other cange need to happen to the classical transformation for this to work, such as the classical christofells connexion which does not work anymore in this context so we introduce the spin(or latin) connexion whit

$$\nabla_\mu X_b^a = \partial_\mu X_b^a + \omega_\mu^a{}_c X_b^c - \omega_\mu^c{}_b X_c^a \quad (8)$$

Of course we can still make link with the levi-civita connexion with the equations

$$\Gamma_{\mu\lambda}^\nu = V_a^\nu \partial_\mu V_\lambda^a + V_a^\nu V_\lambda^b \omega_\mu^a{}_b, \quad (9)$$

$$\omega_\mu^a{}_b = V_\nu^a V_b^\lambda \Gamma_{\mu\lambda}^\nu - V_b^\lambda \partial_\mu V_\lambda^a, \quad (10)$$

With those relation we can do prove the anti-symetrie of the last two index of the spin connexion

$$\begin{aligned}\nabla_\mu \eta_{ab} &= \partial_\mu X_{ab} - \omega_\mu{}^a{}_c \eta_{cb} - \omega_\mu{}^c{}_b \eta_{ac} \\ &= -\omega_{\mu ab} - \omega_{\mu ba} \\ &= 0\end{aligned}\tag{11}$$

so in effect we have

$$-\omega_{\mu ab} = \omega_{\mu ba}\tag{12}$$

2.2 Groupe and gauge transformation

Now we will look at both the gauge liberty of the theory and the symetrie of the $O(E,E)$ group and it's action on different object. so firstly the different gauge invariance we will encounter in this internship are $x'^\mu = x^\mu + \xi^\mu(x)$, and $B_{\mu\nu} \rightarrow B_{\mu\nu} + \partial_\mu u \Lambda_\nu^{(2)}$ for the first one we can refer to the [4] and use the Lie derivative to establish how different object will transform:

$$\delta\phi(x) = \xi^\mu \partial_\mu \Phi(x)\tag{13}$$

$$\delta U^\mu = \xi^\rho \partial_\rho U^\mu - (\partial_\rho \xi^\mu) U^\rho\tag{14}$$

$$\delta\omega_\mu = \xi^\rho \partial_\rho \omega_\mu + (\partial_\rho \xi^\mu) \omega_\rho\tag{15}$$

$$\delta T_\nu^\mu = \xi^\rho \partial_\rho T_\nu^\mu - (\partial_\rho \xi^\mu) T_\nu^\rho + (\partial_\nu \xi^\rho) T_\rho^\mu\tag{16}$$

Now we will look at the action of the $O(E,E)$ group, but firstly, let's define what is the $O(E,E)$ group:

Definition 1 Any matrices M is in $O(E,E)$ if it is a square $E \times E$ matrices and if for the matrices

$$\zeta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

We can use this matrices as a metric for the groupe $O(E,E)$ as it has d -1 eigenvalues and d 1 eigenvalues, in a basis rotated from the one with diagonal metric. the matrices M verify

$$\boxed{M^T \zeta M = \zeta}$$

Now the question is how does this group act on a vector, a tensor form or other things. We have

$$M^{\mu\nu} \rightarrow M^{\mu\nu} + \Psi_\lambda^\mu M^{\lambda\gamma} \Psi_\gamma^\nu\tag{17}$$

$$A^\mu \rightarrow A^\mu + \Psi_\nu^\mu A^\nu\tag{18}$$

3 Lagrangian field theory

3.1 Lagrangian density

In field theory, just as Newton's second law is "volumized" into the Navier-Stokes equations by distributing quantities over space, we replace discrete degrees of freedom with continuous field densities defined at every point in spacetime. Therefore we will rewrite the Lagrangian as the integral over space of a new quantities called Lagrangian density, which will often referred to as just the Lagrangian. Thus we obtain this equation:

$$L = \iiint \mathcal{L}(\eta_\rho, \frac{\partial \eta_\rho}{\partial x^\nu}) dx dy dz\tag{19}$$

Now using esintein convention we can rewrite the action in a new light :

$$S = \int \mathcal{L} dx^\mu\tag{20}$$

3.2 General exemple for scalar and vectorial fields, and result for two forms

In field theory, the Lagrangian has a classical prescription for scalar and vector fields, but no general rule exists for other cases each field type requires its own treatment. For this work, only the scalar, vector, and two-form Lagrangians are needed, along with the metric, which is handled by Einstein's equations.

So for a scalar field we have :

$$\boxed{\mathcal{L} = -\frac{1}{2} \partial_\mu \Phi \partial^\mu \Phi}\tag{21}$$

and for a vectorial field we have

$$\mathcal{L} = -\frac{1}{2}F_{\mu\nu}F^{\mu\nu} \quad (22)$$

where for the field A

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

In a vacuum the einstein equation give us

$$\mathcal{L} = R$$

where R is ricci scalar finally for the lagrangian of a two form fields we will look in [5] where it is given by the equation :

$$\mathcal{L} = \frac{1}{12}g^{\hat{\mu}\hat{\mu}'}g^{\nu\nu'}g^{\rho\rho'}H_{\mu\nu\rho}H_{\mu'\nu'\rho'} \quad (23)$$

where for the two form $B_{\nu\rho}$ we have

$$H_{\mu\nu\rho} = \partial_\mu B_{\nu\rho} + cyc.perm$$

4 NS-NS Action Of supergravity

In this section we perform a Kaluza-Klein reduction. We consider a universe living in $D + E$ dimensions, with $D = 4$, and we try to determine how an observer confined to the D -dimensional world perceives the extra dimensions.

4.1 Hypothesis

We assume that none of the fields depend on the internal coordinates, i.e. for any field $\partial_\alpha A^\mu = 0$. This mean that the reduction will be done on a E-d torus.

4.2 Rewriting the action

We start from the NS-NS action, which is

$$S = \int dx^D \int \frac{dy^E}{\rho(E)} \sqrt{-\hat{g}} e^{\hat{\phi}} (\hat{R} + \hat{g}^{\hat{\mu}\hat{\nu}} \partial_{\hat{\mu}} \hat{\phi} \partial^{\hat{\nu}} \hat{\phi} - \frac{1}{12} g^{\hat{\mu}\hat{\mu}'} g^{\nu\nu'} g^{\rho\rho'} H_{\mu\nu\rho} H_{\mu'\nu'\rho'}) \quad (24)$$

where $\rho(E)$ is a normalisation factor, corresponding in practice to the invariant volume of the internal space.

And where $\hat{R}$ is similar as the classical Hilbert-Einstein action but in $D + E$ dimension, $\hat{\phi}$ is a dilaton field and $-\frac{1}{12}H_{\mu\nu\rho}H^{\mu\nu\rho}$ is the action of a two form. Finally as no field depend on internal component and knowing what $\rho(E)$ we can just simplify this integral. Now we will separate this action into 3 part one for each action so we have

$$S_{\hat{g}} = \int dx^D \sqrt{-\hat{g}} e^{-\hat{\phi}} \hat{R} \quad (25a)$$

$$S_{\hat{\phi}} = \int dx^D \sqrt{-\hat{g}} e^{-\hat{\phi}} \hat{g}^{\hat{\mu}\hat{\nu}} \partial_{\hat{\mu}} \hat{\phi} \partial^{\hat{\nu}} \hat{\phi} \quad (25b)$$

$$S_{\hat{B}} = \int dx^D \sqrt{-\hat{g}} e^{-\hat{\phi}} - \frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho} \quad (25c)$$

Now using the fact that $\hat{R}$ can be written as

$$\hat{R} = \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \hat{R}_{\hat{\mu}\hat{\nu}\hat{r}\hat{s}}$$

where

$$\hat{R}_{\hat{\mu}\hat{\nu}\hat{r}\hat{s}} = \partial_{\hat{\mu}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} + \hat{\omega}_{\hat{\mu}\hat{r}}^{\hat{t}} \hat{\omega}_{\hat{\nu}\hat{t}\hat{s}} - (\mu \leftrightarrow \nu)$$

and the ω are defined By

$$\hat{\omega}_{\hat{t}\hat{r}\hat{s}} = -\hat{\Omega}_{\hat{t}\hat{r}\hat{s}} + \hat{\Omega}_{\hat{r}\hat{s}\hat{t}} - \hat{\Omega}_{\hat{s}\hat{t}\hat{r}}$$

and finally the Ω are defined thanks to the vielbein with

$$\hat{\Omega}_{\hat{r}\hat{s}\hat{t}} = \frac{1}{2} (\hat{V}_{\hat{r}}^{\hat{\mu}} \hat{V}_{\hat{s}}^{\hat{\nu}} - \hat{V}_{\hat{r}}^{\hat{\nu}} \hat{V}_{\hat{s}}^{\hat{\mu}}) \partial_{\hat{\nu}} \hat{V}_{\hat{\mu}\hat{r}}$$

with all of that we are going to rewrite the action $S_{\hat{g}}$ in termes of $\hat{\omega}$

$$\begin{aligned}
S_{\hat{g}} &= \int dx^D \hat{V} e^{-\hat{\phi}} \hat{R} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \hat{R}_{\hat{\mu}\hat{\nu}\hat{r}\hat{s}} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} (\partial_{\hat{\mu}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} + \hat{\omega}_{\hat{\mu}\hat{r}}^{\hat{t}} \omega_{\hat{\nu}\hat{t}\hat{s}} - \partial_{\hat{\nu}} \hat{\omega}_{\hat{\mu}\hat{r}\hat{s}} - \hat{\omega}_{\hat{\nu}\hat{r}}^{\hat{t}} \omega_{\hat{\mu}\hat{t}\hat{s}}) \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} ([\partial_{\hat{\mu}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} - \partial_{\hat{\nu}} \hat{\omega}_{\hat{\mu}\hat{r}\hat{s}}] + [\hat{\omega}_{\hat{\mu}\hat{r}}^{\hat{t}} \omega_{\hat{\nu}\hat{t}\hat{s}} - \hat{\omega}_{\hat{\nu}\hat{r}}^{\hat{t}} \omega_{\hat{\mu}\hat{t}\hat{s}}]) \\
&= \int dx^D \{2\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} + \hat{V} e^{-\hat{\phi}} [\hat{\omega}_{\hat{r}}^{\hat{t}} \omega_{\hat{t}\hat{s}}^{\hat{s}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \omega_{\hat{t}\hat{s}}^{\hat{t}}]\} \\
&= \int dx^D \{ \hat{V} e^{-\hat{\phi}} [\hat{\omega}_{\hat{r}}^{\hat{t}} \omega_{\hat{t}\hat{s}}^{\hat{s}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \omega_{\hat{t}\hat{s}}^{\hat{t}}] + \underbrace{\partial_{\hat{\mu}} (2\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}})}_{\text{Border terme which goes to zero}} - 2\hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} \partial_{\hat{\mu}} (\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}}) \} \\
&= \int dx^D \{ \hat{V} e^{-\hat{\phi}} [\hat{\omega}_{\hat{r}}^{\hat{t}} \omega_{\hat{t}\hat{s}}^{\hat{s}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \omega_{\hat{t}\hat{s}}^{\hat{t}}] - 2\hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} \partial_{\hat{\mu}} (\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}}) \} \\
&= \int dx^D \{ \hat{V} e^{-\hat{\phi}} [\hat{\omega}_{\hat{r}}^{\hat{t}} \omega_{\hat{t}\hat{s}}^{\hat{s}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \omega_{\hat{t}\hat{s}}^{\hat{t}}] - 2\hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} [\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \partial_{\hat{\mu}} \hat{V}^{\hat{\nu}\hat{s}} + \hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}} \hat{V}^{\hat{\mu}\hat{r}} + e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}} \hat{V} + \hat{V} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}} e^{-\hat{\phi}}] \} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \{ [\hat{\omega}_{\hat{r}}^{\hat{t}} \omega_{\hat{t}\hat{s}}^{\hat{s}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \omega_{\hat{t}\hat{s}}^{\hat{t}}] - 2\hat{\omega}_{\hat{\nu}}^{\hat{r}\hat{s}} [\hat{V}_{\hat{r}}^{\hat{\mu}} \hat{V}_{\hat{s}}^{\hat{\nu}} V_{\hat{t}}^{\hat{\lambda}} \partial_{\hat{\mu}} \hat{V}_{\hat{\lambda}}^{\hat{t}} - \hat{V}_{\hat{r}}^{\hat{\mu}} \hat{V}_{\hat{t}}^{\hat{\nu}} \hat{V}_{\hat{s}}^{\hat{\lambda}} \partial_{\hat{\mu}} \hat{V}_{\hat{\lambda}}^{\hat{t}} - \hat{V}_{\hat{s}}^{\hat{\nu}} \hat{V}_{\hat{t}}^{\hat{\mu}} \hat{V}_{\hat{r}}^{\hat{\lambda}} \partial_{\hat{\mu}} \hat{V}_{\hat{\lambda}}^{\hat{t}} - \hat{V}_{\hat{r}}^{\hat{\mu}} \hat{V}_{\hat{s}}^{\hat{\nu}} \partial_{\hat{\mu}} \hat{\phi}] \} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \{ [\hat{\omega}_{\hat{r}}^{\hat{t}} \omega_{\hat{t}\hat{s}}^{\hat{s}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \omega_{\hat{t}\hat{s}}^{\hat{t}}] - 2\hat{\omega}_{\hat{s}}^{\hat{r}\hat{s}} [\hat{V}_{\hat{r}}^{\hat{\mu}} \hat{V}_{\hat{t}}^{\hat{\nu}} \partial_{\hat{\mu}} \hat{V}_{\hat{\lambda}}^{\hat{t}} - \hat{V}_{\hat{t}}^{\hat{\mu}} \hat{V}_{\hat{r}}^{\hat{\nu}} \partial_{\hat{\mu}} \hat{V}_{\hat{\lambda}}^{\hat{t}} - \hat{V}_{\hat{r}}^{\hat{\mu}} \partial_{\hat{\mu}} \hat{\phi}] + 2\hat{\omega}_{\hat{t}}^{\hat{r}\hat{s}} \hat{V}_{\hat{r}}^{\hat{\mu}} \hat{V}_{\hat{s}}^{\hat{\nu}} \partial_{\hat{\mu}} \hat{V}_{\hat{\lambda}}^{\hat{t}} \} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \{ [\hat{\omega}_{\hat{r}}^{\hat{t}} \omega_{\hat{t}\hat{s}}^{\hat{s}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \omega_{\hat{t}\hat{s}}^{\hat{t}}] - 2\hat{\omega}_{\hat{s}}^{\hat{r}\hat{s}} [2\hat{\Omega}_{\hat{t}\hat{r}}^{\hat{t}} - \hat{V}_{\hat{r}}^{\hat{\mu}} \partial_{\hat{\mu}} \hat{\phi}] + 2\hat{\omega}_{\hat{t}}^{\hat{r}\hat{s}} \hat{V}_{\hat{r}}^{\hat{\mu}} \hat{V}_{\hat{s}}^{\hat{\nu}} \partial_{\hat{\mu}} \hat{V}_{\hat{\lambda}}^{\hat{t}} \} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \{ [\hat{\omega}_{\hat{r}}^{\hat{t}} \omega_{\hat{t}\hat{s}}^{\hat{s}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \omega_{\hat{t}\hat{s}}^{\hat{t}}] + 2\hat{\omega}_{\hat{s}}^{\hat{r}\hat{s}} [2\hat{\omega}_{\hat{t}\hat{r}}^{\hat{t}} + \hat{V}_{\hat{r}}^{\hat{\mu}} \partial_{\hat{\mu}} \hat{\phi}] + 2\hat{\omega}_{\hat{t}}^{\hat{r}\hat{s}} \Omega_{\hat{r}\hat{s}}^{\hat{t}} \} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \{ [\hat{\omega}_{\hat{r}}^{\hat{t}} \omega_{\hat{t}\hat{s}}^{\hat{s}} - \hat{\omega}_{\hat{r}}^{\hat{s}} \omega_{\hat{t}\hat{s}}^{\hat{t}}] + 2\hat{\omega}_{\hat{s}}^{\hat{r}\hat{s}} [2\hat{\omega}_{\hat{t}\hat{r}}^{\hat{t}} + \hat{V}_{\hat{r}}^{\hat{\mu}} \partial_{\hat{\mu}} \hat{\phi}] - \hat{\omega}_{\hat{t}}^{\hat{r}\hat{s}} (\omega_{\hat{s}}^{\hat{t}} \hat{r} + \hat{\omega}_{\hat{r}\hat{s}}^{\hat{t}}) \} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \{ \hat{\omega}_{\hat{s}}^{\hat{t}} \omega_{\hat{r}\hat{t}}^{\hat{r}} + \hat{\omega}_{\hat{r}}^{\hat{s}\hat{t}} \omega_{\hat{s}\hat{t}}^{\hat{r}} + 2\hat{\omega}_{\hat{s}}^{\hat{r}\hat{s}} \hat{V}_{\hat{r}}^{\hat{\mu}} \partial_{\hat{\mu}} \hat{\phi} \}
\end{aligned} \tag{26}$$

4.3 Vielbein Anstaz

Now that we have explicitley written the action in terme of either latin connexion and or the two other fields, what we hate to do is to suppose an anstaz for the vielbein. Why the vielbein and not the metric you may ask? Because as stated in [2.1](#), all the information of the metric can be contained within the vielbein, and has we primarily work with them for the action, it is easier to do an anstaz on them. Now we will do the same anzasts as in the [\[1\]](#), however we will consider that γ they use is null since the begining, and we will use the the same parameter number as in [\[5\]](#). Finally thank to the local lorentz invariance we can choose a triangular parentrisation giving us

$$\hat{V}_{\hat{\mu}}^{\hat{r}} = \begin{pmatrix} V_{\hat{\mu}}^{\hat{r}} & A_{\hat{\mu}}^{(1)\beta} \Phi_{\beta}^{\hat{a}} \\ 0 & \Phi_{\hat{\alpha}}^{\hat{a}} \end{pmatrix} \tag{27}$$

Now still using [\(2\)](#) we can compute the inverse vielbein and then the metric and then the inverse metric that gave us

$$\hat{V}_{\hat{\mu}}^{\hat{r}} = \begin{pmatrix} V_{\hat{\mu}}^{\hat{r}} & -A_{\hat{\mu}}^{\hat{\alpha}} \\ 0 & \Phi_{\hat{\alpha}}^{\hat{a}} \end{pmatrix}, \tag{28}$$

for the inverse vielbein, and

$$\hat{g}_{\hat{\mu}\hat{\nu}} = \begin{pmatrix} g_{\mu\nu} - A_{\mu}^{\gamma} A_{\nu\lambda} & -A_{\mu\beta} \\ A_{\nu\alpha} & -h_{\alpha\beta} \end{pmatrix}, \text{ and } \hat{g}^{\hat{\mu}\hat{\nu}} = \begin{pmatrix} g^{\mu\nu} & -A^{\mu\beta} \\ -A^{\nu\alpha} & -h^{\alpha\beta} + A_{\rho}^{\beta} A^{\rho\alpha} \end{pmatrix}, \tag{29}$$

for the metric

Now that we have made this anstaz an intersting things to do is to verify what does object are by using the classical transformation $x'^{\hat{\mu}} = x^{\hat{\mu}} + \xi^{\hat{\mu}}(x)$ and then using what we saw in part [2.2](#) that with the Lie derivate we can rewrote this transformation as being Using the fact that the theory is invariant under the classical transformation $x'^{\hat{\mu}} = x^{\hat{\mu}} + \xi^{\hat{\mu}}(x)$ [\[1\]](#), which can be rewritten with [\(13\)](#) as

$$\delta \hat{V}_{\hat{\mu}}^{\hat{r}} = \xi^{\hat{\rho}} \partial_{\hat{\rho}} \hat{V}_{\hat{\mu}}^{\hat{r}} + \partial_{\hat{\mu}} \xi^{\hat{\rho}} \hat{V}_{\hat{\rho}}^{\hat{r}} \tag{30}$$

we now apply this transformation to each block of the vielbein, in order to see whether it corresponds to some familiar type of field still thanks to (13). Let us start with the block $\hat{V}_\mu^r$:

$$\begin{aligned}\delta\hat{V}_\mu^r &= \xi^\rho \partial_\rho \hat{V}_\mu^r + \partial_\mu \xi^\rho \hat{V}_\rho^r \\ &= \xi^\rho \partial_\rho \hat{V}_\mu^r + \partial_\mu \xi^\nu \hat{V}_\nu^r + \underbrace{\partial_\mu \xi^\alpha \hat{V}_\alpha^r}_{\text{this is null}} \\ &= \xi^\rho \partial_\rho (V_\mu^r + \partial_\mu \xi^\nu V_\nu^r) \\ &= (\xi^\rho \partial_\rho V_\mu^r + \partial_\mu \xi^\nu V_\nu^r) + \xi^\rho \partial_\rho V_\mu^r\end{aligned}$$

We can also redo this computation using a different method:

$$\begin{aligned}\delta\hat{V}_\mu^r &= \delta(V_\mu^r) \\ &= \delta(V_\mu^r)\end{aligned}$$

and by identification we find

$$\delta V_\mu^r = \xi^\rho \partial_\rho V_\mu^r + \partial_\mu \xi^\nu V_\nu^r$$

which is exactly the transformation law of a vector field under a Lie derivative [4]. Applying the same method to the other blocks, we obtain, for Φ_α^a :

$$\delta\Phi_\alpha^a = \xi^\rho \partial_\rho \Phi_\alpha^a$$

which is exactly the transformation law of a scalar field. Finally, for A_μ^α we obtain:

$$A_\mu^\alpha = \xi^\rho \partial_\rho A_\mu^\alpha + \partial_\mu \xi^\rho A_\rho^\alpha + \frac{\partial_\mu \xi^\alpha}{2\kappa}$$

which corresponds to the transformation law of a vector field, up to an extra correction term. One may further note that, restricting to ξ^α only, we obtain

$$\delta V_\mu^r = \delta\Phi_\alpha^a = 0, \quad (31a)$$

$$\delta A_\mu^\alpha = \frac{\partial_\mu \xi^\alpha}{2\kappa}. \quad (31b)$$

so that what remains of the D -dimensional diffeomorphisms, once restricted to ξ^α independent of the internal coordinates y , corresponds to a set of E abelian gauge symmetries. The vielbein in (27) thus contains the D -dimensional vielbein (i.e. the graviton), E vector fields A_μ^α , and E^2 scalar fields Φ_α^a . However, because of the local Lorentz invariance in the internal directions, only $\frac{1}{2}E(E+1)$ of these scalars actually correspond to propagating degrees of freedom. We also introduce $\delta = \det\Phi$

5 Ricci scalar reduction

Now that we better know the tool we will use, We can begin the calculation. For a starter we will do the Einstein action in $D+E$ dimension twiked by the addition of the dilation.

Firstly let's begin by expending completly the $S_{\hat{g}}$, this will show us that we technically have to compute 8 ω , however thanks to the antisymetrie og the last to index we can reduce that to 6. So the action is

$$\begin{aligned}S = -\frac{1}{4\kappa^2} \int d^D x \int \frac{d^E y}{\rho(E)} \hat{V} \Big[&\hat{\omega}^a_{bc} \hat{\omega}^{bc}_a - \hat{\omega}^a_{tb} \hat{\omega}^{bt}_a + \hat{\omega}^a_{sc} \hat{\omega}^{sc}_a + \hat{\omega}^a_{st} \hat{\omega}^{st}_a - \hat{\omega}^r_{bc} \hat{\omega}^{b\ c}_r - \hat{\omega}^r_{tb} \hat{\omega}^{bt}_r - \hat{\omega}^r_{sc} \hat{\omega}^{s\ c}_r \\ &\hat{\omega}^r_{st} \hat{\omega}^{st}_r + \hat{\omega}^a_{ac} \hat{\omega}^{b\ c}_c + \hat{\omega}^a_{ta} \hat{\omega}^{bt}_b + \hat{\omega}^a_{ac} \hat{\omega}^{s\ c}_s - \hat{\omega}^a_{ta} \hat{\omega}^{s\ t}_t + \hat{\omega}^r_{rc} \hat{\omega}^{b\ c}_c - \hat{\omega}^r_{rt} \hat{\omega}^{bt}_b + \hat{\omega}^r_{rc} \hat{\omega}^{s\ c}_s + \hat{\omega}^r_{rt} \hat{\omega}^{s\ t}_t + \\ &2\hat{\omega}_s^{\ r} \hat{V}_r^\mu \partial_\mu \hat{\phi} + 2\hat{\omega}_b^{\ r} \hat{V}_r^\mu \partial_\mu \hat{\phi} \Big] \quad (32)\end{aligned}$$

It then becomes clear that we only need to compute six components of ω , namely

$$\begin{aligned}\hat{\omega}_{pmn}, \hat{\omega}_{pma}, \hat{\omega}_{pab}, \\ \hat{\omega}_{cmn}, \hat{\omega}_{cma}, \hat{\omega}_{cab},\end{aligned}$$

One can also check that computing these six components only requires the following eight components of Ω :

$$\begin{aligned}\hat{\Omega}_{pmn}, \\ \hat{\Omega}_{pma}, \hat{\Omega}_{pam}, \hat{\Omega}_{apm}, \\ \hat{\Omega}_{mab}, \hat{\Omega}_{amb}, \hat{\Omega}_{abm}, \\ \hat{\Omega}_{abc},\end{aligned}$$

Using (28), (4.2) and (4.2), we can now set out a clear strategy: first the Ω 's, and then the ω 's. Now let's begin the computation of the Ω 's. Starting with $\hat{\Omega}_{pmn}$:

$$\begin{aligned}
\hat{\Omega}_{pmn} &= \frac{1}{2}(\hat{V}_p^{\hat{\mu}}\hat{V}_m^{\nu} - \hat{V}_m^{\hat{\mu}}\hat{V}_p^{\nu})\partial_{\nu}\hat{V}_{\hat{\mu}n} \\
&= \frac{1}{2}(\hat{V}_p^{\mu}\hat{V}_m^{\nu} - \hat{V}_m^{\mu}\hat{V}_p^{\nu})\partial_{\nu}\hat{V}_{\mu n} \\
&= \frac{1}{2}(\hat{V}_p^{\mu}\hat{V}_m^{\nu} - \hat{V}_m^{\mu}\hat{V}_p^{\nu})\partial_{\nu}\hat{V}_{\mu n} \\
&= \frac{1}{2}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})\partial_{\nu}(V_{\mu n}) \\
&= \frac{1}{2}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})\partial_{\nu}(V_{\mu n}) \\
&= \frac{1}{2}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})\partial_{\nu}(V_{\mu n}) \\
&= \frac{1}{2}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})\partial_{\nu}(V_{\mu n}) \\
&= \frac{1}{2}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})\partial_{\nu}(V_{\mu n})
\end{aligned}$$

For $\hat{\Omega}_{pma}$, we find:

$$\begin{aligned}
\hat{\Omega}_{pma} &= \frac{1}{2}(\hat{V}_p^{\hat{\mu}}\hat{V}_m^{\nu} - \hat{V}_m^{\hat{\mu}}\hat{V}_p^{\nu})\partial_{\nu}\hat{V}_{\hat{\mu}a} \\
&= -(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})\partial_{\nu}(A_{\mu}^{\alpha}\Phi_{a\alpha}) + (A_p^{\alpha}V_m^{\nu} - \hat{A}_m^{\alpha}V_p^{\nu})\partial_{\nu}(\Phi_{a\alpha}) \\
&= -(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})(\partial_{\nu}(A_{\mu}^{\alpha})\Phi_{a\alpha} + \partial_{\nu}(\Phi_{a\alpha})A_{\mu}^{\alpha}) + \kappa 1(A_p^{\alpha}V_m^{\nu} - \hat{A}_m^{\alpha}V_p^{\nu})\partial_{\nu}(\Phi_{a\alpha}) \\
&= -(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})(\partial_{\nu}(A_{\mu}^{\alpha})\Phi_{a\alpha} + \partial_{\nu}(\Phi_{a\alpha})A_{\mu}^{\alpha}) + (A_p^{\alpha}V_m^{\nu} - \hat{A}_m^{\alpha}V_p^{\nu})\partial_{\nu}(\Phi_{a\alpha}) \\
&= -(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})\partial_{\nu}(A_{\mu}^{\alpha}\Phi_{a\alpha}) \\
&= (F^{\alpha}_{pm}\Phi_{a\alpha} - \partial_{\nu}(V_p^{\mu}V_m^{\nu})A_{\mu}^{\alpha}\Phi_{a\alpha} + \partial_{\nu}(V_p^{\nu}V_m^{\mu})A_{\mu}^{\alpha}\Phi_{a\alpha}) \\
&= F^{\alpha}_{pm}\Phi_{a\alpha}
\end{aligned}$$

For $\hat{\Omega}_{pam}$:

$$\begin{aligned}
\hat{\Omega}_{pam} &= \frac{1}{2}(\hat{V}_p^{\hat{\mu}}\hat{V}_a^{\nu} - \hat{V}_a^{\hat{\mu}}\hat{V}_p^{\nu})\partial_{\nu}\hat{V}_{\hat{\mu}m} \\
&= \frac{1}{2}(\hat{V}_p^{\mu}\hat{V}_a^{\nu} - \hat{V}_a^{\mu}\hat{V}_p^{\nu})\partial_{\nu}\hat{V}_{\mu m} \\
&= 0
\end{aligned}$$

The same result holds for $\hat{\Omega}_{apm}$.

For $\hat{\Omega}_{pab}$:

$$\begin{aligned}
\hat{\Omega}_{pab} &= \frac{1}{2}(\hat{V}_p^{\hat{\mu}}\hat{V}_a^{\nu} - \hat{V}_a^{\hat{\mu}}\hat{V}_p^{\nu})\partial_{\nu}\hat{V}_{\hat{\mu}b} \\
&= \frac{-1}{2}\hat{V}_a^{\hat{\mu}}V_p^{\nu}\partial_{\nu}\hat{V}_{\hat{\mu}b} \\
&= \frac{1}{2}\Phi_a^{\alpha}V_p^{\nu}\partial_{\nu}\Phi_{\alpha b} \\
&= \frac{1}{2}\Phi_a^{\alpha}V_p^{\nu}\partial_{\nu}(h_{\alpha\beta}\Phi_b^{\beta}) \\
&= \frac{1}{2}\Phi_a^{\alpha}\Phi_b^{\beta}V_p^{\nu}\partial_{\nu}h_{\alpha\beta} + \frac{1}{2}\Phi_a^{\alpha}V_p^{\nu}h_{\alpha\beta}\partial_{\nu}\Phi_b^{\beta} \\
&= \frac{1}{2}\Phi_a^{\alpha}\Phi_b^{\beta}V_p^{\nu}\partial_{\nu}h_{\alpha\beta} + \frac{1}{2}\Phi_{\beta a}V_p^{\nu}\partial_{\nu}\Phi_b^{\beta} \\
&= \frac{1}{2}\Phi_a^{\alpha}\Phi_b^{\beta}V_p^{\nu}\partial_{\nu}h_{\alpha\beta} + \frac{1}{2}V_p^{\nu}\partial_{\nu}(\underbrace{\Phi_{\beta a}\Phi_b^{\beta}}_{=\delta_{ab} \text{ i.e is constant}}) - \Phi_b^{\beta}V_p^{\nu}\partial_{\nu}\Phi_{\beta a} \\
&= \frac{1}{2}\Phi_a^{\alpha}\Phi_b^{\beta}V_p^{\nu}\partial_{\nu}h_{\alpha\beta} - \Phi_b^{\beta}V_p^{\nu}\partial_{\nu}\Phi_{\beta a}
\end{aligned}$$

For $\hat{\Omega}_{apb}$:

$$\begin{aligned}\hat{\Omega}_{apb} &= \frac{1}{2}(\hat{V}_a^{\hat{\mu}}\hat{V}_p^{\nu} - \hat{V}_p^{\hat{\mu}}\hat{V}_a^{\nu})\partial_{\nu}\hat{V}_{\hat{\mu}b} \\ &= \frac{1}{2}\hat{V}_a^{\hat{\mu}}\hat{V}_p^{\nu}\partial_{\nu}\hat{V}_{\hat{\mu}b} \\ &= \frac{1}{2}\hat{V}_a^{\alpha}\hat{V}_p^{\nu}\partial_{\nu}\hat{V}_{\alpha b} \\ &= \frac{-1}{2}\Phi_a^{\alpha}V_p^{\nu}\partial_{\nu}\Phi_{\alpha b}\end{aligned}$$

For $\hat{\Omega}_{abp}$:

$$\begin{aligned}\hat{\Omega}_{abp} &= \frac{1}{2}(\hat{V}_a^{\hat{\mu}}\hat{V}_b^{\nu} - \hat{V}_b^{\hat{\mu}}\hat{V}_a^{\nu})\partial_{\nu}\hat{V}_{\hat{\mu}p} \\ &= 0\end{aligned}$$

The same holds for Ω_{abc} . Using (??), we can then obtain the ω 's straightforwardly:

$$\hat{\omega}_{pmn} = \omega_{pmn}, \quad (33a)$$

$$\hat{\omega}_{pma} = F_{pm}^{\alpha}\Phi_{\alpha a}, \quad (33b)$$

$$\hat{\omega}_{pab} = \frac{1}{2}\Phi_a^{\alpha}V_p^{\mu}\partial_{\mu}\Phi_{\alpha\beta} - (a \leftrightarrow b), \quad (33c)$$

$$\hat{\omega}_{cmn} = -F_{mn}^{\alpha}\Phi_{\alpha c}, \quad (33d)$$

$$\hat{\omega}_{cab} = 0 \quad (33e)$$

We can now conclude this part by just computing the multiplication and look at what we got. This give us.

$$\boxed{S = \int dx^D \int V \delta \left(\frac{-1}{4} g^{\rho\lambda} \partial_{\lambda} h_{\alpha\beta} \partial_{\rho} h^{\alpha\beta} + \frac{1}{4} F^{\mu\nu\alpha} F_{\mu\nu}^{\beta} h_{\alpha\beta} + \right.} \quad (34)$$

$$\left. + R + \frac{1}{4} \partial_{\hat{\mu}} \log(h) \partial^{\hat{\mu}} \log(h) + \partial_{\hat{\mu}} \log(h) \partial^{\hat{\mu}} \hat{\phi} \right)$$

We can observe some interesting things. Firtstly we can see that we have refound the classical Einstein action, coupled with E electromagnetism field. So what we can conclude is that gravity in D+E dimation is gravity in 4d plus E electromagnetism field. and we find a few other field which will come in handy for the dilaton field reduction.

5.1 Adding the dilaton field reduction

For this part we will have to sum $S_{\hat{g}}$ and $S_{\hat{\phi}}$ that gave us

$$\begin{aligned}S_{\hat{g}+\hat{\phi}} &= \int dx^D \int V \delta \left(\frac{-1}{4} g^{\rho\lambda} \partial_{\lambda} h_{\alpha\beta} \partial_{\rho} h^{\alpha\beta} + \frac{1}{4} F^{\mu\nu\alpha} F_{\mu\nu}^{\beta} h_{\alpha\beta} + \right. \\ &\quad \left. + R + \frac{1}{4} \partial_{\hat{\mu}} \log(h) \partial^{\hat{\mu}} \log(h) + \partial_{\hat{\mu}} \log(h) \partial^{\hat{\mu}} \hat{\phi} + \partial_{\hat{\mu}} \hat{\phi} \partial^{\hat{\mu}} \hat{\phi} \right)\end{aligned}$$

We then introduce the shifted dilaton as

$$\phi = \hat{\phi} - \frac{1}{2} \log h = \hat{\phi} - \log \delta \quad (35)$$

wich gave us

$$\begin{aligned}S &= \int dx^D V e^{-\phi} \left(R - \frac{1}{4} F^{\mu\nu\alpha} F_{\mu\nu}^{\beta} h_{\alpha\beta} + \right. \\ &\quad \left. - g^{\rho\lambda} \partial_{\lambda} h_{\alpha\beta} \partial_{\rho} h^{\alpha\beta} + \hat{g}^{\hat{\mu}\hat{\nu}} \partial_{\hat{\mu}} \phi \partial_{\hat{\nu}} \phi \right)\end{aligned} \quad (36)$$

6 Two form fiel reduction

6.1 The reduction

For the two form we will reduce it apart from the two other field, this those not cause any problem as the action is in effect linear.

We can rewrite (23) as

$$\begin{aligned}
\mathcal{L}_{\hat{B}} &= \frac{-1}{12} (g^{\hat{\mu}\hat{\mu}'} g^{\hat{\nu}\hat{\nu}'} g^{\hat{\rho}\hat{\rho}'} \hat{H}_{\hat{\mu}\hat{\nu}\hat{\rho}} \hat{H}_{\hat{\mu}'\hat{\nu}'\hat{\rho}'}) \\
&= \frac{-1}{12} (\hat{\eta}^{\hat{r}\hat{r}'} \hat{\eta}^{\hat{s}\hat{s}'} \hat{\eta}^{\hat{t}\hat{t}'} \hat{H}_{\hat{r}\hat{s}\hat{t}} \hat{H}_{\hat{r}'\hat{s}'\hat{t}'}) \\
&= \frac{-1}{12} (\hat{H}_{\hat{r}\hat{s}\hat{t}} \hat{H}^{\hat{r}\hat{s}\hat{t}}) \\
&= \frac{-1}{12} (\hat{H}_{rst} \hat{H}^{rst} + \hat{H}_{rsc} \hat{H}^{rsc} + \hat{H}_{rbt} \hat{H}^{rbt} + \hat{H}_{rbc} \hat{H}^{rbc} + \hat{H}_{ast} \hat{H}^{ast} + \hat{H}_{asc} \hat{H}^{asc} + \hat{H}_{abs} \hat{H}^{abs} + \hat{H}_{abc} \hat{H}^{abc})
\end{aligned}$$

however thanks to the fact that $\hat{B}_{\hat{\mu}\hat{\nu}}$ is antisymmetric one can easily show that this imply that $\hat{H}_{\hat{\mu}\hat{\nu}\hat{\rho}}$ is antisymmetric for any neighbor index. so we obtain

$$\mathcal{L}_{\hat{B}} = \frac{-1}{12} (\hat{H}_{rst} \hat{H}^{rst} + 3\hat{H}_{rsc} \hat{H}^{rsc} + 3\hat{H}_{asc} \hat{H}^{asc} + \hat{H}_{abc} \hat{H}^{abc}) \quad (37)$$

and finally the action of B is written as

$$S_{\hat{b}} = - \int dx^D \sqrt{-g} e^{-\phi} \left\{ \frac{1}{12} H_{\alpha\beta\gamma} H^{\alpha\beta\gamma st} + \frac{1}{4} H_{\mu\alpha\beta} H^{\mu\alpha\beta} + \frac{1}{4} H_{\mu\nu\alpha} H^{\mu\nu\alpha} + \frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho} \right\} \quad (38)$$

The absence of hat is due to what we saw in equation (7)

so with that we obtain (38)

Now we will compute each component of non-hatted H in function of hatted H

to begin slowly as we have seen above $\hat{B}_{\alpha\beta} = B_{\alpha\beta}$ and if we use the y-independence we get $H_{\alpha\beta\gamma} = 0$

We continue with $H_{\mu\alpha\beta}$

$$\begin{aligned}
H_{\mu\alpha\beta} &= V_{\mu}^r \hat{V}_r^{\hat{\mu}} \hat{H}_{\hat{\mu}\alpha\beta} \\
&= \partial_{\mu} B_{\alpha\beta}
\end{aligned} \quad (39)$$

those two were quite easy however the next two are going to be more troublesome, but we are going to introduce new objects

Now for $H_{\mu\nu\rho}$

$$\begin{aligned}
H_{\mu\alpha\beta} &= V_{\mu}^r V_{\nu}^s \hat{V}_r^{\hat{\mu}} \hat{V}_s^{\hat{\nu}} \hat{H}_{\hat{\mu}\hat{\nu}\alpha} \\
&= \hat{H}_{\mu\nu\alpha} - (A_{\mu}^{(1)\alpha} \hat{H}_{\alpha\nu\rho} + A_{\rho}^{(1)\alpha} \hat{H}_{\mu\alpha\nu}) \\
&= \partial_{\mu} \hat{B}_{\nu\alpha} + \partial_{\nu} \hat{B}_{\mu\alpha} - A_{\mu}^{(1)\beta} \partial_{\nu} \hat{B}_{\alpha\beta} - A_{\nu}^{(1)\beta} \partial_{\mu} \hat{B}_{\beta\alpha} \\
&= \partial_{\mu} \hat{B}_{\nu\alpha} + \partial_{\nu} \hat{B}_{\mu\alpha} - \partial_{\nu} (A_{\mu}^{(1)\beta} \hat{B}_{\alpha\beta}) + \partial_{\nu} A_{\mu}^{(1)\beta} \hat{B}_{\alpha\beta} - \partial_{\mu} (A_{\nu}^{(1)\beta} \hat{B}_{\beta\alpha}) + \partial_{\mu} A_{\nu}^{(1)\beta} \hat{B}_{\beta\alpha} \\
&= \partial_{\mu} \hat{B}_{\nu\alpha} - \partial_{\nu} \hat{B}_{\mu\alpha} - \partial_{\nu} (A_{\mu}^{(1)\beta} \hat{B}_{\alpha\beta}) + \partial_{\nu} A_{\mu}^{(1)\beta} \hat{B}_{\alpha\beta} + \partial_{\mu} (A_{\nu}^{(1)\beta} \hat{B}_{\beta\alpha}) - \partial_{\mu} A_{\nu}^{(1)\beta} \hat{B}_{\beta\alpha} \\
&= \partial_{\mu} (\hat{B}_{\nu\alpha} + A_{\nu}^{(1)\beta} \hat{B}_{\alpha\beta}) - \partial_{\nu} (\hat{B}_{\mu\alpha} + A_{\mu}^{(1)\beta} \hat{B}_{\alpha\beta}) - \hat{B}_{\alpha\beta} (\partial_{\mu} A_{\nu}^{(1)\beta} - \partial_{\nu} A_{\mu}^{(1)\beta}) \\
&= F_{\mu\nu\alpha}^{(2)} - \hat{B}_{\alpha\beta} F_{\mu\nu}^{(1)\beta}
\end{aligned} \quad (40)$$

and finally for $H_{\mu\nu\rho}$ we are just going to use the exact same technique as before which gave us

$$\begin{aligned}
H_{\mu\nu\rho} &= V_{\mu}^r V_{\nu}^s V_{\rho}^t \hat{V}_r^{\hat{\mu}} \hat{V}_s^{\hat{\nu}} \hat{V}_t^{\hat{\rho}} \hat{H}_{\hat{\mu}\hat{\nu}\hat{\rho}} \\
&= \hat{H}_{\mu\nu\rho} - A_{\mu}^{(1)\alpha} \hat{H}_{\alpha\nu\rho} - A_{\nu}^{(1)\alpha} \hat{H}_{\alpha\rho\mu} - A_{\rho}^{(1)\alpha} \hat{H}_{\alpha\mu\nu} + A_{\mu}^{(1)\alpha} A_{\nu}^{(1)\beta} \hat{H}_{\alpha\beta\rho} + A_{\nu}^{(1)\alpha} A_{\rho}^{(1)\beta} \hat{H}_{\alpha\beta\mu} + A_{\rho}^{(1)\alpha} A_{\mu}^{(1)\beta} \hat{H}_{\alpha\beta\nu} \\
&= \partial_{\mu} B_{\nu\rho} - \frac{1}{2} (A_{\mu}^{(1)\alpha} F_{\nu\rho\alpha}^{(2)} + A_{\mu\alpha}^{(2)} F_{\nu\rho}^{(1)}) + \partial_{\nu} B_{\rho\mu} - \frac{1}{2} (A_{\nu}^{(1)\alpha} F_{\rho\mu\alpha}^{(2)} + A_{\nu\alpha}^{(2)} F_{\rho\mu}^{(1)}) + \partial_{\rho} B_{\mu\nu} - \frac{1}{2} (A_{\rho}^{(1)\alpha} F_{\mu\nu\alpha}^{(2)} + A_{\rho\alpha}^{(2)} F_{\mu\nu}^{(1)})
\end{aligned} \quad (41)$$

where

$$B_{\mu\nu} = \hat{B}_{\mu\nu} + \frac{1}{2} A_{\mu}^{(1)\alpha} A_{\nu\alpha}^{(2)} - \frac{1}{2} A_{\nu}^{(1)\alpha} A_{\mu\alpha}^{(2)} - A_{\mu}^{(1)\alpha} B_{\alpha\beta} A_{\nu}^{(2)\beta} \quad (42)$$

6.2 Observation of gauge

Now as in part 4.3 we are going to observe the gauge rule to be sure that every object transforms as it should so let's begin with $A^{(2)}$ which should transform with the Lie derivative and under the new transformations of $B_{\mu\nu} \rightarrow B_{\mu\nu} + \partial_{\mu} u \Lambda_{\nu}$, so let's begin with $A_{\mu\alpha}^{(2)}$

$$\begin{aligned}
\delta A_{\mu\alpha}^{(2)} &= \delta (\hat{B}_{\mu\alpha} + B_{\alpha\beta} A_{\mu}^{(1)\beta}) \\
&= \delta (\hat{B}_{\mu\alpha}) + \delta (B_{\alpha\beta} A_{\mu}^{(1)\beta}) \\
&= \delta (\hat{B}_{\mu\alpha}) + \delta (B_{\alpha\beta}) A_{\mu}^{(1)\beta} + B_{\alpha\beta} \delta (A_{\mu}^{(1)\beta}) \\
&= L_{\xi} \hat{B}_{\mu\alpha} + \partial_{\mu} \Lambda_{\alpha}^{(2)} + A_{\mu}^{(\beta)} L_{\xi} \hat{B}_{\alpha\beta} + \hat{B}_{\alpha\beta} \partial_{\mu} \Lambda_{\alpha}^{(1)} \\
&= L_{\xi} \hat{B}_{\mu\alpha} + \partial_{\mu} \Lambda_{\alpha}^{(2)} + L_{\xi} (A_{\mu}^{(\beta)} \hat{B}_{\alpha\beta}) \\
&= \partial_{\mu} \Lambda_{\alpha}^{(2)} + L_{\xi} A_{\mu\alpha}^{(2)}
\end{aligned} \quad (43)$$

so we have the right transformation for this object

Doing the same thing with $B_{\mu\nu}$ we come under the transformation

$$\delta B_{\mu\nu} = \frac{1}{2}(\Lambda^{(1)\alpha} F_{\mu\nu\alpha}^{(2)} + \lambda_\alpha^{(2)} F_{\mu\nu}^{(1)\alpha}) \quad (44)$$

which is exactly the good term to compensate the additional term for the transformation of $H_{\mu\nu\rho}$

7 Complete reduced action

To recap we have the complete reduced action in the form

$$S = \int dx^D \sqrt{-g} e^{-\phi} \mathcal{L} \quad (45)$$

Where we decompose $\mathcal{L}$ as $\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_2 + \mathcal{L}_3 + \mathcal{L}_4$, and we pose

$$\begin{aligned} \mathcal{L}_1 &= R + g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi \\ \mathcal{L}_2 &= \frac{1}{4} g^{\mu\nu} (\partial_\mu h_{\alpha\beta} \partial_\nu G^{\alpha\beta} - h^{\alpha\beta} h^{\gamma\delta} \partial_\mu B_{\alpha\gamma} \partial_\nu B_{\beta\delta}) \\ \mathcal{L}_3 &= -\frac{1}{4} g^{\mu\rho} g^{\nu\lambda} (h_{\alpha\beta} F_{\mu\nu}^{(1)\alpha} F_{\rho\lambda}^{(1)\beta} + h^{\alpha\beta} H_{\mu\nu\alpha} H_{\rho\lambda\beta}) \\ \mathcal{L}_4 &= -\frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho} \end{aligned} \quad (46)$$

8 symmetries observation

Now for this part we are going to explore a very important part of physics research nowadays which is symmetry. In our context it will allow us to make equivalence between multiple theories as a whole. But firstly we have to see the obvious symmetry that we have seen in the non-reduced version of the Action which is the $SL(E, R)$. But what will interest us in the reduction is that it does not create nor destroy any symmetry and if we observe a new symmetry in the reduced form that means that we have effectively discovered a new symmetry in the unreduced form. and following the [5] we observe an $O(E, E)$ symmetry (defined here ??).

We will now define a new matrix M which is

$$M = \begin{pmatrix} h^{-1} & -h^{-1}B \\ Bh^{-1} & h - Bh^{-1}B \end{pmatrix} \quad (47)$$

Of course we will verify that this is true

$$\begin{aligned} M^T \zeta M &= \begin{pmatrix} h^{-1} & Bh^{-1} \\ -h^{-1}B & h - Bh^{-1}B \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} h^{-1} & -h^{-1}B \\ Bh^{-1} & h - Bh^{-1}B \end{pmatrix} \\ &= \begin{pmatrix} Bh^{-1} & h^{-1} \\ h - Bh^{-1}B & -h^{-1}B \end{pmatrix} \begin{pmatrix} h^{-1} & -h^{-1}B \\ Bh^{-1} & h - Bh^{-1}B \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \zeta \end{aligned}$$

so we confirm that $M \in O(E, E)$ Now that we have that in mind let's retake (46) and verify that each $\mathcal{L}$ is in fact $O(E, E)$ invariant

For $\mathcal{L}_1$ the invariance is obvious as $g_{\mu\nu}$ and Φ are both invariant. Nonwithstanding we ought to remember that for Φ each term in it is not on its own self-invariant only.

Now the [5] propose a new tactic to observe the invariance of $\mathcal{L}_2$ which is to rewrite everything in matrixial terms. we then obtain

$$\mathcal{L}_2 = \frac{1}{4} \text{tr}(\partial_\mu h^{-1} \partial^\mu h + h^{-1} \partial_\mu B h^{-1} \partial^\mu B)$$

And now we are going to do a few manipulations

$$\begin{aligned} \mathcal{L}_2 &= \frac{1}{4} \text{tr}(\partial_\mu h^{-1} \partial^\mu h + h^{-1} \partial_\mu B h^{-1} \partial^\mu B) \\ &= \frac{1}{8} \text{tr} \left(2 \partial_\mu h \partial^\mu h^{-1} + 2 h^{-1} \partial_\mu B h^{-1} \partial^\mu B \right) \\ &= \frac{1}{8} \text{tr} \left(2 \partial_\mu h \partial^\mu h^{-1} - 2 \partial_\mu (B h^{-1} B) \partial^\mu h^{-1} + \partial_\mu (B h^{-1}) \partial^\mu (B h^{-1}) + \partial_\mu (h^{-1} B) \partial^\mu (h^{-1} B) \right) \\ &= \frac{1}{8} \text{tr} \left(\partial_\mu (h - B h^{-1} B) \partial^\mu h^{-1} + \partial_\mu (B h^{-1}) \partial^\mu (B h^{-1}) + \partial_\mu (h^{-1} B) \partial^\mu (h^{-1} B) + \partial^\mu h^{-1} \partial_\mu (h - B h^{-1} B) \right) \\ &= \frac{1}{8} \text{tr}(\partial_\mu M^{-1} \partial^\mu M) \end{aligned} \quad (48)$$

Now we are going to rewrite $\mathcal{L}_3$ by developing the H partial to get

$$\begin{aligned}\mathcal{L}_3 &= -\frac{1}{4}[F_{\mu\nu}^{(1)\alpha}h_{\alpha\beta}F^{(1)\mu\nu\beta} + (F_{\mu\nu}^{(2)} - B_{\alpha\gamma}F_{\mu\nu}^{(1)\gamma})h^{\alpha\beta}(F_{\beta}^{(2)\mu\nu} - B_{\beta\delta}F^{(1)\mu\nu\delta})] \\ &= -\frac{1}{4}\mathcal{F}_{\mu\nu}^i(M^{-1})_{ij}\mathcal{F}^{\mu\nu j}\end{aligned}\quad (49)$$

Where $\mathcal{F}_{\mu\nu}^i$ is the 2E-component vector of field strengths

$$\mathcal{F}_{\mu\nu}^i = \begin{pmatrix} F_{\mu\nu}^{(1)\alpha} \\ F_{\mu\nu}^{(2)} \end{pmatrix} = \partial_\mu \mathcal{A}_\nu^i - \partial_\nu \mathcal{A}_\mu^i \quad (50)$$

Now the O(E,E) invariance is visible if we consider that the vector fields transform linealy to the vector representation of O(E,E), i.e., $\mathcal{A}_\mu^i \rightarrow \Omega_j^i \mathcal{A}_\mu^j$. And finally we go to $\mathcal{L}_4$ firstly let's rewrite $H_{\mu\nu\rho}$

$$\begin{aligned}H_{\mu\nu\rho} &= \partial_\mu B_{\nu\rho} - \frac{1}{2}(A_\mu^{(1)\alpha}F_{\nu\rho\alpha}^{(2)} + A_{\mu\alpha}^{(2)}F_{\nu\rho}^{(1)}) + \partial_\nu B_{\rho\mu} - \frac{1}{2}(A_\nu^{(1)\alpha}F_{\rho\mu\alpha}^{(2)} + A_{\nu\alpha}^{(2)}F_{\rho\mu}^{(1)}) + \partial_\rho B_{\mu\nu} - \frac{1}{2}(A_\rho^{(1)\alpha}F_{\mu\nu\alpha}^{(2)} + A_{\rho\alpha}^{(2)}F_{\mu\nu}^{(1)}) \\ &= \partial_\mu B_{\nu\rho} - \frac{1}{2}\mathcal{A}_\mu^i \zeta_{ij} \mathcal{F}_{\nu\rho}^j + (cyc.perms)\end{aligned}\quad (51)$$

Now as $\Omega^T \zeta \Omega = \zeta$ the second term is O(E,E) invariant. And the first term is invariant if we require that $B_{\nu\rho}$ does not transform. This demonstrates that the action hides a non-trivial symmetry, providing a useful framework for comparing equivalent theories.

9 Conclusion

References

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A lagrangian

By using the principle of least action and the variational principle we can see that

$$\begin{aligned}\delta S &= \delta \int \mathcal{L} dx^\mu \\ &= \delta \int \frac{\partial \mathcal{L}}{\partial \eta_\rho} \delta \eta_\rho + \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} \delta \partial_\nu \eta_\rho dx^\mu \\ &= \delta \int \frac{\partial \mathcal{L}}{\partial \eta_\rho} \delta \eta_\rho - \partial_\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} \delta \eta_\rho + \underbrace{\partial_\nu \left(\frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} \delta \eta_\rho \right)}_{\text{border terme equal to zero}} dx^\mu \\ &= 0\end{aligned}$$

which gave the equation

$$\boxed{\frac{\partial \mathcal{L}}{\partial \eta_\rho} - \partial_\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} = 0} \quad (52)$$

which is in fact the equation of mouvement for the field

B General relativity 101

Definition 2 Let V be an R -vector space of dimension N , we define the tensor of order $(r, s) \in \llbracket 0, 1 \rrbracket^2$ on $V^{(r,s)} \stackrel{\text{def}}{=} V^{\otimes r} \otimes V^{\otimes s}$ as a being a multilinear application on $V^{\star r} \times V^s$ to $\mathbb{R}$

Definition 3 We will call n -dimension ($n \in \mathbb{N}^*$) manifold every Hausdorff topological space $(\mathcal{M}, \tau)$ locally homeomorphic to $\mathbb{R}^n$

Definition 4 Let $(\mathcal{M}, g)$ be a smooth manifold. A metric tensor is a symmetric, non-degenerate bilinear form $g_{\mu\nu}$ on the tangent space $T_p\mathcal{M}$ at each point $p \in \mathcal{M}$, such that

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

In General Relativity we use the signature $(-, +, +, +)$.

Definition 5 In general in RG we use the christoffels connexion as it is the sole one wich is mtreci and torsion free. We define it as being

$$\text{nabla}_\mu V^\nu = \partial_\mu V^\nu + \Gamma_{\mu\sigma}^\nu V^\sigma \quad (53)$$

where the christoffels symbols are

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}) \quad (54)$$